MA1521 L1 Week 11 Mon 20261026

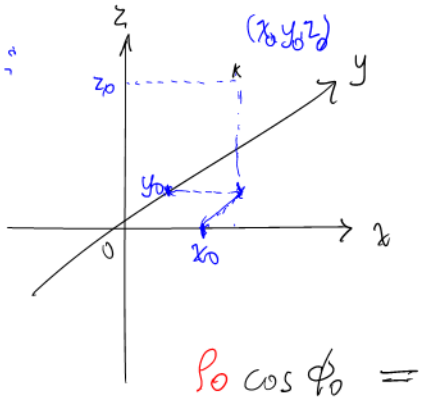
Supply limits for iterated integrals in spherical coordinates $\iiint d\rho\, d\phi\, d\theta$ for each of the following regions. (No integrand is specified; $d\rho\, d\phi\, d\theta$ is given so as to determine the order of integration.)

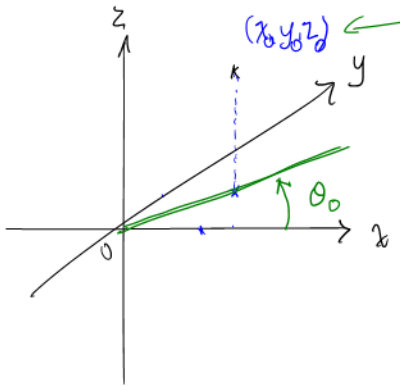
a) The region bounded below by the cone $z^2 = x^2 + y^2$, and above by the sphere of radius $\sqrt{2}$ and center at the origin.

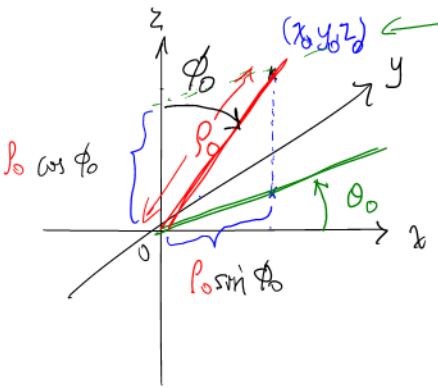
b) The first octant.

c) That part of the sphere of radius 1 and center at $z = 1$ on the $z$-axis which lies above the plane $z = 1$.

Spherical coords:



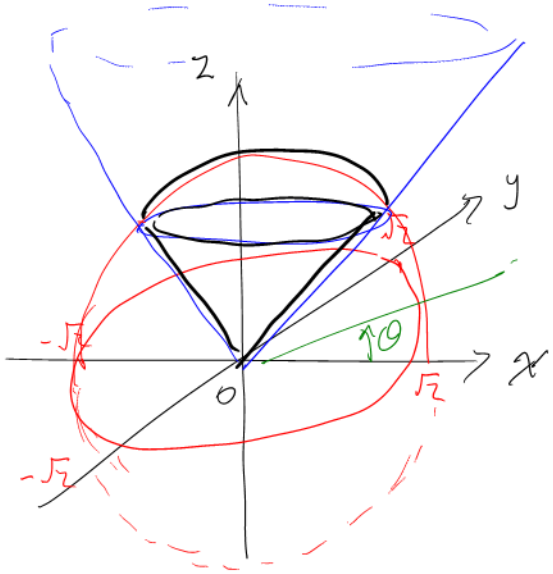

$$\rho_0 \cos\phi_0 = z_0$$

$$\left.\begin{aligned} \rho_0 \sin\phi_0 \cdot \cos\theta_0 &= x_0 \\ \rho_0 \sin\phi_0 \cdot \sin\theta_0 &= y_0 \end{aligned}\right\} \quad \rho_0 \sin\phi_0 \longleftrightarrow r_0 \text{ in polar coords}$$

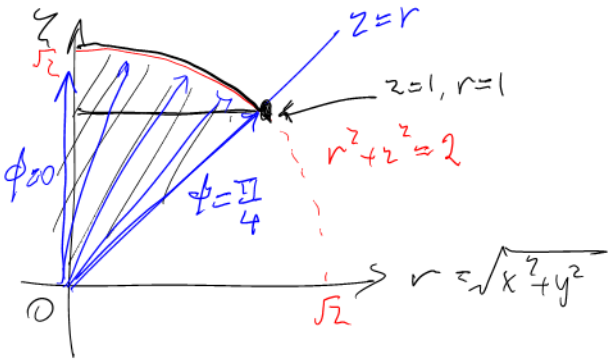
Key formula: $\boxed{dx\,dy\,dz = \rho^2 \sin\phi\; d\rho\, d\phi\, d\theta}$

a) The region bounded below by (the cone $z^2 = x^2 + y^2$,) and above by (the sphere of radius $\sqrt{2}$ and center at the origin.)

$r^2$ $\qquad$ $x^2+y^2+z^2=2$, $r^2$



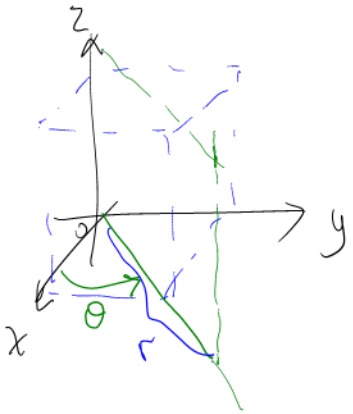

Fix $\theta \in [0, 2\pi]$



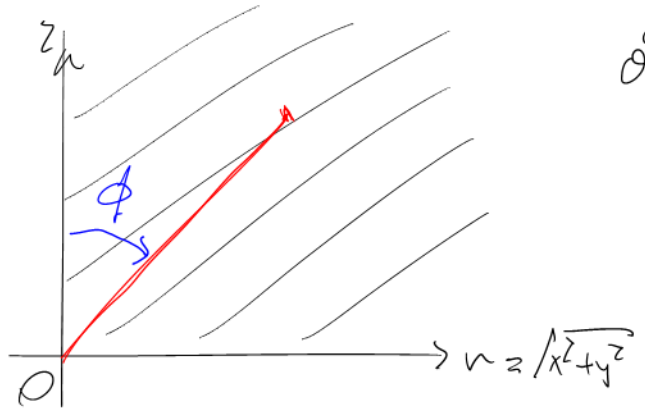

$\theta = 2\pi$ $\quad \phi = \frac{\pi}{4}$ $\quad \rho = \sqrt{2}$

$\int_{\theta=0} \int_{\phi=0} \int_{\rho=0}$ $\quad d\rho\, d\phi\, d\theta$

b) The first octant. $\quad x \geq 0$, $y \geq 0$, $z \geq 0$




Fix $\theta \in [0, \frac{\pi}{2}]$



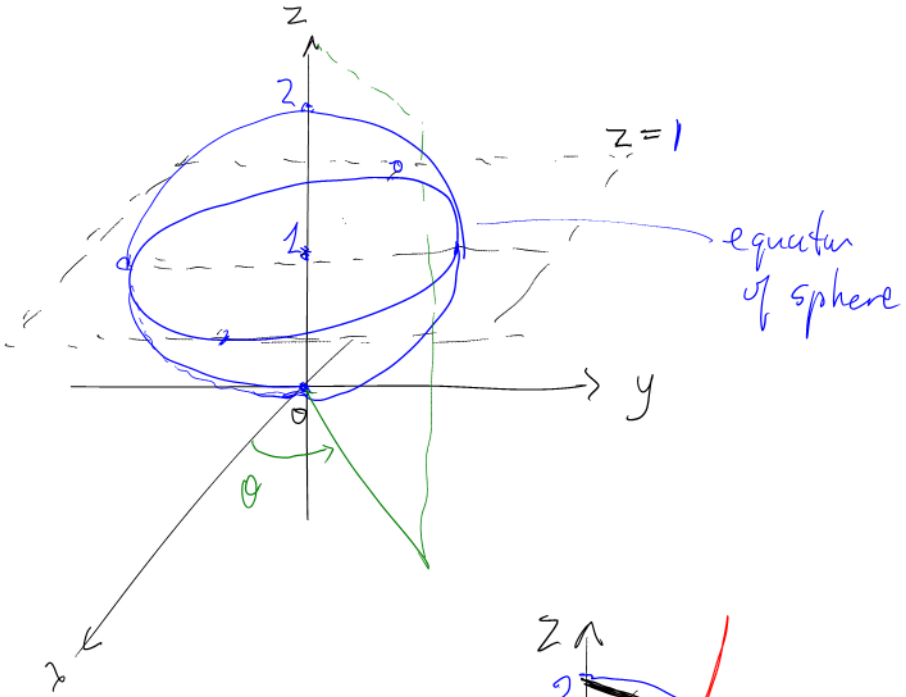

$\theta = \frac{\pi}{2}$ $\quad \phi = \frac{\pi}{2}$ $\quad \rho = \infty$

$\int_{\theta=0} \int_{\phi=0} \int_{\rho=0}$ $\quad d\rho\, d\phi\, d\theta$

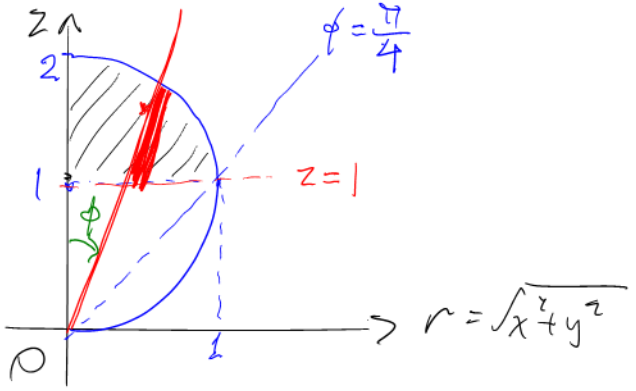
"northern hemisphere"

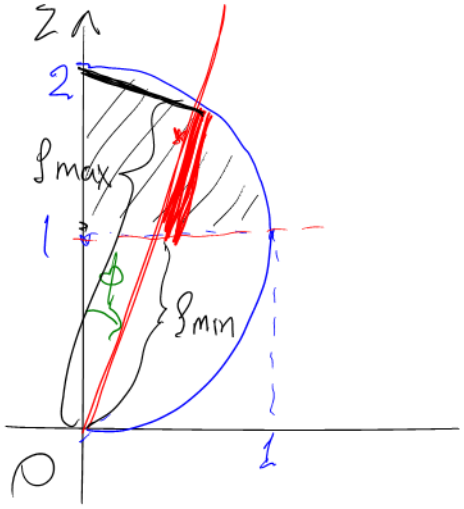
c) That part of the sphere of radius 1 and center at $z = 1$ on the $z$-axis which lies above the plane $z = 1$.

$x=0,\ y=0$ $\qquad z \geq 1$




Fix $\theta \in [0, 2\pi]$




$$\int_{\theta=0}^{2\pi} \int_{\phi=0}^{\frac{\pi}{4}} \int_{\rho=\frac{1}{\cos\phi}}^{2\cos\phi} d\rho\, d\phi\, d\theta$$

$\rho_{min} \cos\phi = 1 \quad \Rightarrow \quad \rho_{min} = \frac{1}{\cos\phi}$

$\rho_{max} = 2\cos\phi$

Find the center of mass of a hemisphere $R$ of radius $a$, using spherical coordinates. Assume the density $\delta = 1$.

Center of mass

$$\left( \frac{\iiint_R x \, dV}{\iiint_R 1 \, dV}, \frac{\iiint_R y \, dV}{\iiint_R 1 \, dV}, \frac{\iiint_R z \, dV}{\iiint_R 1 \, dV} \right)$$

$\iiint_R 1 \, dV$ : Vol.

**Evaluate:** $\iiint_W (x^2 + y^2 + z^2)^{-1/2}\, dx\, dy\, dz$ where $W$ is the region determined by $\frac{1}{2} \leq z \leq 1$ and $x^2 + y^2 + z^2 \leq 1$.

(a previous question, but now use spherical coordinates instead)

**Evaluate:** $\iiint_W \exp(x^2 + y^2 + z^2)^{3/2}\, dV$ where $W$ is the unit ball in $\mathbb{R}^3$.